Categories
All Categories
Batteries & Chargers
Bearings
Building Materials
Connectors & Switches
Electronics
Merchandise
Gift Certificates
Kits & Parts
Motors & Mounts
Pulleys & Belts
Radio Equipment
Servos & Mounts
Tools & Fasteners
Weapon Systems
Wheels & Hubs
Clearance

Shopping Cart
Shopping cart
0 Product(s) in cart
Total 0.00CAD
» Checkout

Your Account
Login Status
Not logged in
» Login

Store Links
Store Search
Safety
Pulley/Belt Calculator
Distributors
Sponsorships
About Us
Jobs
Contact Us
FAQ

VISA Verified by VISA MasterCard SecureCode. PayPal

Home » Categories » Motors & Mounts



FingerTech "Silver Spark" 16mm Gearr

Price: 27.03CAD

19.09 **USD** 16.96 **EUR**

★★★★☆

4.4 stars, based on 25 review(s)

The Silver Spark line of motors has the same 16mm diameter as the Gold are a larger pitch. This means less worrying about stripping teeth in your applications!

• Extra-long 3mm shaft for mounting wheels directly with the option to use support on the wheel's far side.

• Same 11mm face mount pattern as Gold Spark motors.

• All-metal spur gearhead mated to a Mabuchi FK-050 motor.

• 8 different gear ratios for a wide range of rpm and torque to choose from one twice as slow as the Gold Sparks!

• Weights from 28 - 32grams (0.99 - 1.13oz). (see details)

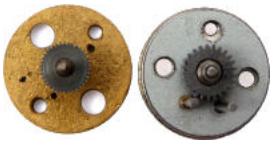
Gear Ratio Please Select...

- 1 + Add to cart

Description | CAD Files | Reviews | Related

April 2026: We've added the 50:1 ratio to go along with the 22:1's and 33:1's for Viper-length pre-cut shafts! Look for the "-short" in the d

The Silver Spark line of motors has the same diameter gearhead as the Gold Sparks, but the gears are a larger pitch. This means less v stripping teeth in your bigger or heavier applications!



(Comparing Gold Spark output gear to Silver Spark)

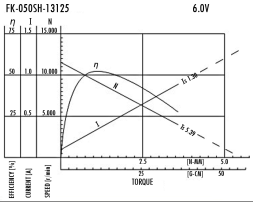
- Extra-long 3mm shaft for mounting wheels directly with the option to use a bearing block for support on the wheel's far side.
- Same 11mm face mount pattern as Gold Spark motors.

- All-metal spur gearhead mated to a Mabuchi FK-050 motor.
- 9 different gear ratios: 11.1, 22.2, 33.3, 50, 83.3 (discontinued, dwindling stock), 100, 200, 300, and 600 :1. One twice as fast, and one the Gold Sparks!
- Weight between 28 - 32grams (0.99 - 1.13oz).

These motors make it super simple to get a robot moving! Just pick up a pair of Foam Wheels and Twist Hubs to mount them, solder on and your motors are ready to be driven by radio control! For mounting the motors, use 2-56 screws and our custom motor mounts. We 3mm bearing blocks for supporting the shaft in heavier robots or weapon systems, and Flat Motor Mounts for when you want to face-mo

Note - For strenuous conditions (such as robot combat), we recommend removing the gearbox and checking that the motor vent hole dc a gearbox pin. If it does, simply rotate the gearbox 180 degrees and reattach. Use Loctite on the screws that hold the gearbox to the mo they cannot separate during competition!

Specifications:

Motor (Mabuchi FK-050SH-13125) Specifications	**Physical Specifications**
Operating Voltage: 3V - 18.5V (>10V decreases motor life) Nominal Voltage: 6V, but most commonly run at 11.1V. (14.8V runs fine but support the shaft because the gears are taking more abuse.) No Load RPM: 11530rpm No Load Current: 0.09A Stall Current: 1.3A Stall Torque: 0.0461kg-cm (0.64oz-in) Kt: 0.0352kg-cm/A (0.49oz-in/A) Kv: 1573rpm/V Peak Efficiency: 52% RPM @ Peak Eff: 9100 Current @ Peak Eff: 0.34A Click herefor the motor torque, current, rpm and efficiency curve. 	Gearbox Lengths: ratios 11: 11.5mm (0.45in) ratios 22, 33, 50: 12.9mm (0.51in) ratios 83, 100: 14.3mm (0.56in) ratios 200, 300: 16.4mm (0.65in) ratios 600: 17.9mm (0.70in) Motor Length: 28.4mm (1.12in) Motor weight (no gearbox): 17g (0.60oz) Gearbox Diameter: 16mm (0.63in) Shaft Diameter: 3mm (0.12in) with flat along length Shaft Length: 38mm (1.5in), and now also 22mm (0.87in) 50:1 ratios for Viper spares Mounting Holes (2): #2-56 spaced 11mm (0.433in) apart -do not use screws that protrude more than 2mm (0.078in Weight: - ratios 11: 28g (0.99oz) - ratios 22, 33, 50, 83, 100: 29g (1.02oz) - ratios 200, 300: 31g (1.09oz) - ratios 600: 32g (1.13oz)

All ratio gearboxes can be interchanged because they use the same 14T pinion gears on the motor.
So (for example) you can remove the 50:1 ratio gearbox from its motor and replace it with a 33:1 ratio gearbox for more speed - without desolder the motor! (Might want to swap the stickers so you know what's in the robot!)

No Load Speed @ Gear Ratio/Voltage

Speed (rpm)	Voltage						
Gear ratio	5	6	7.4	11.1	14.8	18.5	22.2
11.1 :1	709	850	1049	1573	2097	2622	3146
22.2 :1	354	425	524	786	1049	1311	1573
33.3 :1	236	283	350	524	699	874	1049
50 :1	157	189	233	349	466	582	698
83.3 :1	94	113	140	210	279	349	419
100 :1	79	94	116	175	233	291	349
200 :1	39	47	58	87	116	146	175
300 :1	26	31	39	58	78	97	116
600 :1	13	16	19	29	39	49	58

(Click to Enlarge)

Motors have been tested to work up to 22.2V (6 lipoly cells), but don't expect the motor to run that high forever!

Maximum Torque @ Ratio/Voltage (kg-cm) (oz-

Torque (kg-cm)	Voltage						
Gear ratio	5	6	7.4	11.1	14.8	18.5	22.2
11.1 :1	0.35	0.42	0.52	0.77	1.03	1.29	1.55
22.2 :1	0.70	0.84	1.03	1.55	2.07	2.58	3.10
33.3 :1	1.05	1.26	1.55	2.32	3.10	3.87	4.65
50 :1	1.57	1.89	2.33	3.49	4.65	5.82	6.98
83.3 :1	2.62	3.14	3.88	5.81	7.75	9.69	11.63
100 :1	3.14	3.77	4.65	6.98	9.30	11.63	13.96
200 :1	6.29	7.54	9.30	13.96	18.61	23.26	27.91
300 :1	9.43	11.32	13.96	20.93	27.91	34.89	41.87
600 :1	18.86	22.63	27.91	41.87	55.82	69.78	83.74

(Click to Enlarge)

Exceeding 9.72kg-cm (135oz-in) will damage the output gear stage. Do not stall the motors indicated voltage.

Ask A Question Email Friend

Product Highlights:



FingerTech Brushless Mega Spark - Planetary Gearmotor 1500KV



PETG Sheet 1/16"



FingerTech 'Viper' Horizontal Spinner Add-On



FingerTech S3M Timing Pulley 30T for F4006 motor



FingerTech 1lb Viper Parts Bundle



FingerTe